Patent Infringement Analysis: US 10,484,528 B2 (Samsung) — Cross-Device Gesture-Triggered Continuity

TL;DR

- **The single product most exposed to literal infringement of US 10,484,528 B2 is Apple's HomePod/HomePod mini "Handoff" feature** (iPhone-tapped-on-HomePod transfers the currently-playing song, podcast, or call). It cleanly maps to every limitation of Claim 1: a tap/hover gesture performed in proximity with a second device; automatic Bluetooth/UWB link; receipt of context about the audio currently playing; execution of the corresponding media app; performance of the playback function. **HIGH severity / High confidence.**
- Other notably exposed products — all rated **HIGH** or **MEDIUM-HIGH** — are **Apple iOS 17+ AirDrop/NameDrop proximity sharing and SharePlay-by-proximity**, **Huawei Share OneHop / Multi-Screen Collaboration**, **Xiaomi HyperConnect "贴贴分享" Touch-to-Share + App Relay**, **OPPO/OnePlus/realme tap-share via O+ Connect / ColorOS Touch-to-Share**, and the **Google "Tap to Share" feature being built into Android 17, Google Play Services, and Samsung One UI 9** (development-stage as of early 2026). Apple's classic Handoff (iPhone↔Mac↔iPad) and Microsoft's Phone Link / Cross-Device Resume sit at **MEDIUM/LOW** because the trigger is ambient proximity or a share-sheet tap, not an affirmative gesture performed in proximity with the receiving device.
- **The patent faces material validity headwinds.** Apple Continuity/Handoff (announced WWDC June 2014, shipped Fall 2014), Android Beam (Android 4.0 Ice Cream Sandwich, October 19, 2011), the "Bump" cross-device gesture app (Bump Technologies, acquired by Google September 16, 2013, discontinued December 31, 2013), and Samsung's own S-Beam (Galaxy S III, 2012) all predate the January 30, 2017 priority date and disclose nearly every step of Claim 1 with the arguable exception of certain dependent-claim limitations. Any enforcement campaign should expect §102/§103 challenges and possibly §101 abstract-idea attacks. Samsung itself almost certainly does not assert this patent because doing so would expose Samsung Flow, Samsung Quick Share / "Tap to share" in One UI 9, and Samsung DeX continuity to invalidity counterclaims. Crunchbase + 2

Key Findings

1. **The asserted claim language is "gesture performed in proximity with the second electronic device"** (verbatim from the patent's abstract, sourced via Justia search-result snippet). This is broader than NFC tap-only and arguably covers hover (BLE/UWB), bump, and physical-tap implementations. The patent uses "first electronic device" / "second electronic device" terminology consistent with Claims 1 (method), 6 (apparatus) and 11 (CRM) being the three independent claims.
2. **Apple's HomePod mini Handoff** uses the U1 ultra-wideband chip plus a deliberate "bring iPhone close" / "tap top of HomePod" gesture to (a) establish a UWB/BLE link, (b) receive

metadata about what is currently playing, (c) auto-launch the relevant media app on the receiving device, and (d) continue playback. AppleInsider, "Testing the new HomePod mini music Handoff feature in iOS 14.4," confirmed on December 17, 2020 that the iOS 14.4/HomePod 14.4 beta "uses the U1 chip of the HomePod mini to accurately track the location of other U1 devices — our iPhone for example." 9to5Mac's December 14, 2022 report quotes Apple's iOS 16.3 user-guide text verbatim: "Bring iPhone close to HomePod to view controls, or when playing music, to move music between iPhone and HomePod." AppleInsider 9to5Mac

3. **Apple's iOS 17 AirDrop proximity / NameDrop / SharePlay-by-proximity** is an NFC-triggered cross-device gesture controlled by an Apple Settings toggle named "Bringing Devices Together" (Settings → General → AirDrop). MacRumors (Sep 2023) confirms a transfer "without you having to tap on the Share Sheet" after the gesture, mapping closely to Claim 1's gesture-triggered link + content transfer + auto-launch.
4. **Huawei Share OneHop / Multi-Screen Collaboration** is the cleanest non-Apple read on Claim 1: a phone tapped against an NFC tag on a Huawei laptop establishes a Bluetooth/Wi-Fi link, transfers the currently-open file or screen-recording context, launches the corresponding app on the receiving device, and performs the function "associated with [the] activity executed in the second device while the gesture is performed" (Huawei consumer support page, consumer.huawei.com/en/support/huaweishare/).
5. **Xiaomi HyperConnect "贴贴分享" (Touch-to-Share)** explicitly states (hyperos.mi.com/continuity/abilities): "Bring close to the NFC area of another Xiaomi phone or iPhone for quick sharing." Combined with Xiaomi's "应用接力" (App Relay) on HyperOS 2/3, the Xiaomi stack covers both gesture-based file/context transfer and gesture-initiated app handoff. MI
6. **OPPO / OnePlus / realme** ship a near-identical capability through O+ Connect (App Store launch March 27, 2025; ColorOS 15+ native integration). The infringement read is weaker for share-sheet-driven flows but cleaner for newer tap-NFC variants.
7. **Google's "Tap to Share"** (NFC-triggered Quick Share) is a development-stage feature found in APK teardowns of Google Play Services and Samsung One UI 9. Per Android Authority's March 30, 2026 exclusive, One UI 9 code includes the UI description **"Just hold the top of your phone close to the device, and the files will be sent,"** plus verbatim XML strings `<string name="tap_to_share_summary">Tap your phone with someone</string>`, `<string name="tap_to_share_sending_dialog_message_requesting_to">Requesting to %1$s</string>`, and `<string name="tap_to_share_sending_dialog_message_sent_to">Sent to %1$s</string>`. When this ships in Android 17 it will read on nearly every limitation of Claim 1.
8. **Samsung is the patent owner — Samsung cannot infringe its own patent.** Samsung Flow, Galaxy Watch call switching, Quick Share, DeX continuity, and One UI 9 "Tap to share" are listed below only because the user asked for completeness; they are corroborating evidence that this is exactly the technology Samsung's patent covers, not infringement.
9. **Automotive call-transfer (CarPlay / Android Auto / Uconnect)** is *not* a strong read because the transfer is triggered by ignition state, USB unplug, or Bluetooth handoff — not by a

user-performed gesture in proximity with the second device. We rate these **NONE/LOW**.

10. **Validity exposure is significant.** Android 4.0 Ice Cream Sandwich (which introduced Android Beam) was announced October 19, 2011 at a Samsung event in Hong Kong alongside the Galaxy Nexus. Bump Technologies was acquired by Google September 16, 2013, with the Bump app discontinued December 31, 2013. Apple Continuity/Handoff was demonstrated at WWDC June 2, 2014 and shipped Fall 2014. All predate the January 30, 2017 priority and disclose the gesture+proximity+context-transfer combination. Wikipedia

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Claim Breakdown Matrix — US 10,484,528 B2

Limitation (paraphrased from Claim 1)	Required element	Notes
[1a] Detect a *gesture* performed in *proximity* with a *second electronic device*	gesture + proximity	Includes hover, tap, bump per spec examples (smartphone-over-smartwatch, hover smartwatch direction over HUD)
[1b] Establish a *communication link* with the second device in response	BLE/NFC/Wi-Fi/UWB link triggered *by* the gesture	Distinguishes from always-on Bluetooth pairing
[1c] Receive *information associated with content* being provided by the second device	Context/metadata (e.g., now-playing track, open news article, active call)	Goes beyond raw file transfer
[1d] Execute an *application* on the first device based on that information	First device auto-launches matching app	"Smart" handoff, not just a download
[1e] Perform a *function* associated with the *activity executed in the second device while the gesture is performed*	App on first device resumes/continues the activity	Temporal coupling — activity must be live on second device at the gesture moment
Claim 2	Communication link establishment via gesture	Narrowing of [1b]
Claim 3	Software keyboard + message app + arrival notification	Messaging-specific embodiment
Claim 4	Related-application correspondence between devices	E.g., Samsung Messages ↔ Google Messages mapping

Limitation (paraphrased from Claim 1)	Required element	Notes
Claim 5	Transmit control information to a third electronic device	E.g., phone tells smartwatch or HUD to do something
Claim 6	Apparatus (electronic device) — same scope as Claim 1	
Claim 11	Non-transitory CRM — same scope as Claim 1	
Claim 13	Identifying function corresponding to activity	Narrowing of [1e]
Claim 14	Different gestures trigger different functions	E.g., short tap = transfer, long hover = mirror
Claim 15	Function performed *automatically* in response to gesture	No additional confirmation step

Details — Per-Product Infringement Reads

1. Apple — HomePod / HomePod mini Handoff (iPhone-to-HomePod music transfer)

Description. User holds an iPhone (11 or later, U1-equipped) close to a HomePod or HomePod mini (HomePod mini and 2nd-gen HomePod include U1). The iPhone provides haptic feedback that intensifies with proximity; on contact-range proximity, the currently-playing audio (Apple Music, Spotify, podcast, FaceTime call) transitions to the speaker (or vice versa), with the corresponding control card auto-appearing.

Timeline. The original HomePod launched February 9, 2018 but Handoff was NOT available at launch. Tap-to-Handoff was first added in HomePod Software 13.2, released October 28, 2019 (MacRumors, Oct 28, 2019: "Handoff music, podcasts or phone calls by bringing your iPhone close to HomePod"). The U1-enhanced gesture-based Handoff arrived in HomePod Software 14.4, announced in beta on December 17, 2020 (AppleInsider). Apple Support

Evidence sources.

- AppleInsider, "Testing the new HomePod mini music Handoff feature in iOS 14.4," Dec 17, 2020.
- 9to5Mac, "iOS 16.3 guides users on using Handoff from iPhone to HomePod," Dec 14, 2022.
- Engadget, "HomePod mini update adds UWB-enhanced handoff features," Dec 17, 2020.
- MacRumors, "How to Disable HomePod Mini Handoff," 2023.

Claim-by-claim.

- [1a] **YES** — bringing/tapping iPhone proximate to HomePod is the gesture; U1 measures angle/proximity.
- [1b] **YES** — BLE link upgrades to UWB-assisted handoff session triggered by proximity.
- [1c] **YES** — first device (iPhone OR HomePod, bi-directional) receives metadata about the now-playing content.
- [1d] **YES** — receiving side launches/resumes the appropriate audio app to continue playback.
- [1e] **YES** — the function (playback of the specific song at the specific timestamp) is *the* activity live on the source device at the moment of the gesture.

Severity: HIGH. Confidence: High. Doctrine of Equivalents not even needed; literal read is clean. Apple may argue the "gesture" is just "proximity," not a *performed* gesture, but the '528 spec expressly calls out "hovering smartphone over smartwatch" as a gesture.

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2. Apple — iOS 17 AirDrop Proximity Sharing / NameDrop / SharePlay-by-proximity

Description. Bringing two iPhones (XS or later, NFC) physically close (top edges meeting) triggers contact card exchange (NameDrop), file/photo transfer (AirDrop proximity), or synchronized SharePlay session start.

Evidence sources.

- MacRumors, "iOS 17 AirDrop Features: NameDrop, SharePlay, and Proximity Sharing," 2023.
- iDownloadBlog, "How to turn off all NFC proximity features on your iPhone," Oct 23, 2023.
- Apple Settings → General → AirDrop → "Bringing Devices Together" toggle.

Claim-by-claim.

- [1a] YES — NFC-mediated proximity bump.
- [1b] YES — link established by the gesture.
- [1c] YES — content metadata (contact card / file / media being watched) transmitted.
- [1d] YES — Contacts app / Photos / SharePlay session opens on receiver. MacRumors
- [1e] PARTIAL/YES — for SharePlay-by-proximity the activity on the source device (e.g., watching a movie) is mirrored to the receiver; for NameDrop the "activity" is viewing one's contact card, a narrower read.

Severity: HIGH for SharePlay-by-proximity; MEDIUM-HIGH for NameDrop / AirDrop proximity. Confidence: High.

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3. Apple — Classic Handoff (iPhone ↔ Mac ↔ iPad ↔ Apple Watch)

Description. Cross-device app continuation: start a Mail draft on iPhone, see the Handoff icon in the Mac Dock; click to continue. **Evidence sources.**

- Apple Support, "Use Handoff to continue tasks on your other Apple devices" (support.apple.com/en-us/102426).
- apple.com/macos/continuity/.

Claim-by-claim.

- [1a] **WEAK/NO** — no affirmative gesture is required; mere proximity + an in-app NSUserActivity broadcast suffices. The user *clicks* the Handoff icon on the receiving device; there is no proximity-detected gesture *performed* with the second device.
- [1b] PARTIAL — BLE proximity check happens passively, not in response to a gesture.
- [1c] YES — NSUserActivity payload transmitted.
- [1d] YES — corresponding app launched.
- [1e] YES — task continues.

Severity: MEDIUM (DoE only) / LOW (literal). Confidence: Medium. Apple's strongest non-infringement argument is the absence of a "gesture performed in proximity with" the second device — only ambient proximity is detected.

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4. Apple — Apple Watch Handoff & Auto-Unlock Mac

Description. Apple Watch can hand off a task to iPhone; iPhone-handoff-from-Watch surfaces an icon. Auto-Unlock unlocks a Mac when the wearer of an authenticated Apple Watch is in proximity.

Severity: LOW. No discrete user gesture; same reasoning as #3.

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5. Huawei — Huawei Share OneHop & Multi-Screen Collaboration

Description. Tap the NFC area on the back of a Huawei phone against the Huawei Share sensor on a Huawei laptop. The phone vibrates; depending on the current activity (photo open, document open, or shake-then-tap for screen recording), different functions are triggered: image transfer, doc transfer, screen mirror, or recording of the laptop's screen on the phone. HUAWEI

Evidence sources.

- consumer.huawei.com/en/support/huaweishare/ — official Huawei support page.
- consumer.huawei.com/en/support/content/en-us15950759/ — "Drag to Transfer Files Between Your Phone and Laptop."

- XDA Forums, "Ubiquitous Tap-to-Connect Functionality, with HUAWEI OneHop Kit," 2022.

Claim-by-claim.

- [1a] YES — NFC-area tap is the gesture; shake + tap is a *different* gesture (read on Claim 14).
- [1b] YES — Bluetooth/Wi-Fi link established.
- [1c] YES — phone receives or transmits metadata about which app/file is open.
- [1d] YES — corresponding app (Image viewer / WPS Office / Multi-Screen window) launched.
- [1e] YES — function (file transfer, screen recording, call switchover) is exactly the activity live at gesture time.
- Claim 14 (different gestures → different functions) reads literally on shake-then-tap vs. tap.

Severity: HIGH. Confidence: High — but US market exposure is limited because Huawei has been effectively excluded from US consumer device sales since 2019. Damages base is small in the US.

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6. Xiaomi — HyperConnect "Touch-to-Share" / 贴贴分享 and App Relay (应用接力)

Description. Touch-to-Share: bring a Xiaomi phone close to another Xiaomi phone or an iPhone (with Xiaomi Interconnectivity app) at the NFC area to share files, contacts, browser data, web pages, Wi-Fi credentials. App Relay: use an app on one device, continue on another. Ships with HyperOS 2/3 on Xiaomi, Redmi, and POCO devices.

Evidence sources.

- hyperos.mi.com/continuity/abilities — Xiaomi official site.
- Xiaomi Interconnectivity (App Store changelog) — "Touch to share – Now supports sharing Wi-Fi from a Xiaomi phone to an iPhone, as well as sharing contacts, browser data, and web pages between devices." App Store
- xiaomisale.com blog, "Xiaomi HyperConnect," 2025.

Severity: HIGH. Confidence: High. US market exposure limited (Xiaomi is not a meaningful US carrier brand).

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7. OPPO / OnePlus / realme — O+ Connect / Multi-Screen Connect / ColorOS Touch-to-Share

Description. O+ Connect launched on the App Store on March 27, 2025 (per Baidu Baike, "officially launched on the Apple App Store on March 27, 2025"), supporting cross-platform file transfer, notification/call/message sync from iPhone to OPPO/OnePlus/realme, and remote Mac control. Newer ColorOS versions add native AirDrop-style interoperability via Quick Share. BaiduWiki + 3

Evidence sources.

- connect.oppo.com — official site.
- oppo.com/en/newsroom/stories/oppo-connect/ — OPPO Connect/Multi-Screen Connect overview.
- App Store listing for "O+ Connect," Guangdong Heytap Technology Corp.

Claim-by-claim. O+ Connect's typical workflow is share-sheet-driven rather than gesture-driven, so [1a] is weaker. Where ColorOS exposes a tap-to-share NFC mode (newer devices, parallel to the Google "Tap to Share" rollout), the read becomes clean.

Severity: MEDIUM-HIGH for tap-NFC variants; LOW-MEDIUM for share-sheet-only variants. Confidence: Medium. US market exposure for OnePlus is meaningful (carrier sales via T-Mobile historically); OPPO/realme effectively absent from US.

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8. Google / Android — Tap to Share (Quick Share NFC) — in development for Android 17

Description. Code strings in Google Play Services and Samsung One UI 9 read "Tap your phone with someone," "Requesting to [device]," "Sent to [device]." A glow animation indicates connection; instruction is to "overlap the tops of both devices with the screens facing up." NFC initiates the handshake; Wi-Fi Direct/Bluetooth handle the bulk transfer. Android Authority Android Headlines

Evidence sources.

- Android Authority exclusive (Mar 30, 2026, via androidheadlines.com and Gadget Hacks coverage): One UI 9's Quick Share contains the UI text **"Just hold the top of your phone close to the device, and the files will be sent,"** plus verbatim XML strings <string name="tap_to_share_summary">Tap your phone with someone</string>, <string name="tap_to_share_sending_dialog_message_requesting_to">Requesting to %1$s</string>, and <string name="tap_to_share_sending_dialog_message_sent_to">Sent to %1$s</string>. Gadget Hacks
- Gadget Hacks, "Android Tap to Share Feature: What the Code Signals," Mar 30, 2026.
- Ubergizmo and TechRepublic Apr 2026 follow-ups.

Severity (forward-looking): HIGH if launched as designed. Confidence: Medium-High. Reads cleanly on the patent — but Samsung is the upstream OEM debuting it in One UI 9 *and* the patent owner.

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9. Microsoft — Phone Link / Link to Windows / Cross-Device Resume / Edge "Continue on PC"

Description. Microsoft's Phone Link mirrors notifications, makes/receives calls, runs Android apps on PC. Cross-Device Resume (Feb 2026, expanded list) supports Samsung, Honor, Huawei, OPPO, vivo, Xiaomi. Edge "Continue on PC" sends a webpage from Edge mobile to a paired PC. Microsoft + 2

Evidence sources.

- Microsoft Support, "Phone Link requirements and setup."
- microsoft.com/en-us/windows/sync-across-your-devices.
- abit.ee, "Cross-Device Resume in Windows 11," 2026.
- blogs.windows.com, "Continue on PC in the Microsoft Edge mobile app," Dec 4, 2017.

Claim-by-claim. [1a] **NO** — the trigger is a *share-sheet tap* on the phone, not a proximity gesture performed with the PC. There is no proximity detection. The phone need not even be in the same room.

Severity: LOW (no gesture/proximity element). Confidence: High.

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10. Vivo — EasyShare, Vivo Office Kit

Description. EasyShare is largely a Wi-Fi Direct file-transfer app (no required proximity gesture). Vivo Office Kit (OriginOS 6, 2025) supports task handoff between vivo phones and iPads. MWM

Severity: LOW-MEDIUM. Confidence: Medium.

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11. Samsung (own products — non-infringement noted, included per task)

- **Samsung Flow** — Bluetooth/Wi-Fi pairing between Galaxy phone and Windows PC or Galaxy tablet; "Handover" function explicitly transfers content/activity. Reads on Claim 1 *except* that the trigger is usually a paired auto-connect, not a discrete proximity gesture. Samsung cannot infringe. Samsung Samsung
- **Samsung Quick Share (and the upcoming One UI 9 "Tap to share")** — same Tap-to-Share NFC mechanism described in section 8. Samsung cannot infringe its own patent.
- **Galaxy Watch handoff / call transfer** — call switching from watch to phone is a UI-button action ("Switch to phone" on Wear OS Galaxy Watches), not a proximity gesture; weaker read. Samsung
- **Samsung DeX** — wireless DeX uses Miracast and is not gesture-triggered.

The strong correspondence between Samsung's own products and the '528 patent is *exactly why* Samsung has not asserted this patent: doing so would invite both invalidity counterclaims and license-on-license countersuits.

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12. Smart Home & Automotive (low priority)

- **Apple CarPlay / Android Auto call transfer.** Call moves from car speakers to phone when ignition is off or USB unplugged — not a user-performed gesture. **NONE.** Apple Community
- **Tesla phone-as-key.** UWB/BLE proximity unlock; no app/content handoff. **NONE.**
- **Google Home / Amazon Alexa handoff.** Voice-driven, not gesture-driven. **NONE.**
- **BMW / Mercedes / Audi infotainment.** Generally Bluetooth-paired call transfer, no gesture. **NONE.**

Summary Ranking Table

Rank	Company	Product / Feature	Severity	Confidence	US Market Exposure	Nc
1	Apple	HomePod / HomePod mini Handoff (U1-based)	**HIGH**	High	Very high — ~15M units sold globally at 2021 peak; ~9.5M in 2023 and ~6.2M in 2025 per Business of Apps / Statista	Cl lite
2	Apple	iOS 17+ AirDrop proximity / SharePlay-by-proximity	**HIGH**	High	Very high	Na sli na rea
3	Huawei	Huawei Share OneHop / Multi-Screen Collaboration	**HIGH**	High	Low (US restrictions)	St cla sm da ba
4	Xiaomi	HyperConnect Touch-to-Share (贴贴分享) + App Relay	**HIGH**	High	Low (limited US distribution)	

Rank	Company	Product / Feature	Severity	Confidence	US Market Exposure	No
5	Google / Samsung	Android 17 "Tap to Share" (forward-looking)	**HIGH (when launched)**	Medium-High	Will be very high	Sti rel of 20
6	OPPO/OnePlus/realme	O+ Connect / ColorOS Touch-to-Share	**MEDIUM-HIGH**	Medium	Low-Medium (OnePlus carrier sales)	
7	Apple	Classic Handoff (iPhone↔Mac↔iPad)	**MEDIUM** (DoE)	Medium	Very high	No ge on pro
8	vivo	Vivo Office Kit (OriginOS 6 task handoff)	**LOW-MEDIUM**	Medium	Very low	
9	Apple	Apple Watch Handoff / Auto-Unlock	**LOW**	Medium	High	No dis ge
10	Microsoft	Phone Link / Cross-Device Resume / "Continue on PC"	**LOW**	High	Very high	No ge pro
11	CarPlay/Android Auto	Call transfer	**NONE**	High	Very high	No ge dri
12	Tesla/BMW/Mercedes/Audi	Phone-as-key, call handoff	**NONE**	High	Very high	
—	Samsung	Samsung Flow / Quick Share / DeX / Galaxy Watch	n/a (patent owner)	n/a	Very high	Ca inf

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Validity Concerns

The patent's January 30, 2017 priority date is its biggest problem. Significant prior art includes:

- **Apple Continuity / Handoff** — publicly demonstrated at WWDC June 2, 2014; shipped in iOS 8 / OS X Yosemite Fall 2014. Discloses cross-device app context transfer over BLE + Wi-Fi based on device proximity. While Handoff lacks an affirmative "gesture," many of its dependent embodiments (e.g., AirDrop's "bump" pattern, iPhone-near-Mac unlock) cover gesture/proximity combinations.
- **Android Beam** — shipped Android 4.0 Ice Cream Sandwich, announced October 19, 2011 at a Samsung event in Hong Kong alongside the Galaxy Nexus. NFC-triggered tap between two phones to transfer the currently-open content (webpage URL, YouTube video, contact card, map location), with the receiving app auto-launched. This is arguably a §102 anticipation reference for Claim 1. Greenlivingpedia XDA Developers
- **Bump app** (Bump Technologies, founded 2009; acquired by Google September 16, 2013 and discontinued December 31, 2013) — bump two phones together to exchange contacts/photos. Discloses the gesture-as-pairing-trigger concept.
- **Samsung S-Beam** (Galaxy S III, 2012) — NFC-initiated, Wi-Fi-Direct-completed cross-device file/content transfer. This is Samsung's own prior art and creates an awkward record for Samsung if it ever asserted this patent.
- **Apple's HomePod Handoff** itself first shipped in HomePod Software 13.2 on October 28, 2019, with U1-enhanced gesture-based Handoff arriving in HomePod Software 14.4 (announced in beta December 17, 2020). Both ship dates are *after* the '528 priority date (Jan 30, 2017) but *before* the issue date (Nov 19, 2019) and the U1 features post-date both — Apple's strongest non-infringement-related defense is therefore independent development, not prior art.

§101 abstract-idea risk is also material: at its core, Claim 1 is "do a thing on a phone when the user bumps it on another phone" — a generic ordered combination of conventional steps. A defendant could file an *Alice* motion at the pleading stage.

Geographic Considerations

US 10,484,528 B2 is a US patent. Enforcement requires acts of infringement *in the United States* (making, using, selling, offering for sale, or importing). Practical implications:

- **Apple** (HomePod, iOS 17 AirDrop, Handoff) — very large US damages base; designed and sold in the US; ITC importation jurisdiction also available.
- **Google** (Android 17 Tap to Share) — strong US presence; software distribution via Play Services makes US use ubiquitous.
- **Huawei** — minimal US consumer device sales since 2019 US restrictions; limited damages base.
- **Xiaomi, OPPO, realme, vivo** — minimal direct US consumer device sales; some accessories via Amazon/eBay; very small US damages base.
- **OnePlus** — carrier sales via T-Mobile historically; modest US damages base.

- **Microsoft** (Phone Link) — pre-installed on essentially every Windows 11 PC sold in the US; very large user base, but the infringement read is weak.

Foreign counterparts of the '528 patent (the priority is Indian provisional 201741003400) would need to be checked separately for ex-US enforcement (likely an Indian and possibly Korean counterpart exist; not confirmed in this analysis).

Recommendations

Stage 1 — Pre-litigation diligence (immediate).

1. Pull the verbatim issued claim text of US 10,484,528 B2 from USPTO Patent Center and cross-check whether the issued claims use "information associated with content" / "function associated with an activity" (as the task brief states) or the abstract's "user intent associated with a context" language. This will materially change the infringement read.
2. Order a freedom-to-operate / prior-art search emphasizing pre-Jan-30-2017 references: WWDC 2014 Continuity sessions, Android Beam documentation, Bump Technologies patents (US 9,438,294 family), Samsung S-Beam documentation.
3. Check PAIR/PTAB for any inter partes review history (none located in this analysis, but verify).
4. Check whether the patent is part of any active Samsung licensing program or has been transferred to a non-practicing entity.

Stage 2 — If asserting (Samsung as plaintiff or assignee).

1. **Primary target: Apple HomePod Handoff.** Cleanest read, biggest damages base, Apple has not designed around. File in EDTX or WDTX with claim-construction emphasis on "gesture in proximity" = "hover/bring/tap."
2. **Secondary target: Google's Android 17 "Tap to Share"** once launched (likely H2 2026). Wait for general availability before filing to avoid ripeness issues.
3. **Do not assert against Apple's classic Handoff** — too vulnerable to the "no gesture" non-infringement defense and to invalidity from Apple's own prior 2014 Continuity disclosures.

Stage 3 — If defending (potential target).

1. **Prior-art defense first.** Build §102/§103 case from Android Beam (announced October 19, 2011) and Samsung S-Beam (2012). For Apple, add WWDC 2014 Continuity disclosure.
2. **§101 motion** at the pleading stage — Claim 1 is highly abstract.
3. **Non-infringement: "gesture" element.** Argue that mere proximity (BLE auto-detect) is not a "gesture performed in proximity with" the second device under proper claim construction.
4. **IPR petition** within one year of service.

Benchmarks that would change these recommendations:

- If the verbatim issued claim text turns out to require an explicit *user-performed* gesture (e.g., "in response to detecting a *user-input* gesture"), then targets in rows 7–10 (classic Handoff, Phone Link) drop to **NONE**, and only NFC-tap and UWB-tap features remain credible reads.
- If a prior-art search surfaces a pre-2017 publication that combines NFC-tap + active-app-context transfer + auto-launch (likely from the Android Beam ecosystem), the patent's validity drops below the threshold worth asserting; recommendation switches to defensive licensing only.
- If Samsung has previously licensed this patent (cross-license to Apple, Google, etc.) as part of a portfolio settlement, all of the above is moot.

Caveats

- **Patent infringement is ultimately a legal determination for the courts.** This memo is a technology/business analysis, not a legal opinion. Claim construction (Markman) frequently changes infringement reads dramatically.
- **The verbatim issued claim text was not obtained in this research.** Direct access to Google Patents and USPTO image-ppubs returned permission errors; analysis is based on the patent's abstract language (Justia snippet: "gesture performed in proximity with the second electronic device" / "user intent associated with a context") and the task brief's paraphrase of the claims. If the issued claim language differs from the brief, severity ratings may shift.
- **Some products may have been redesigned or design-arounds may have been implemented.** Apple, Google, and Microsoft routinely refactor cross-device APIs across OS versions.
- **The patent appears unlitigated** as of this research; no PTAB, ITC, or district-court proceedings involving US 10,484,528 B2 were found, but this is not a substitute for a Docket Navigator / PACER / PTAB Trials search.
- **Samsung cannot infringe its own patent** but its products were analyzed per the user's request.
- **Validity exposure is real.** Cross-device interaction has dense prior art going back at least to 2009 (Bump Technologies founding) and arguably to IrDA-era beaming in the 1990s. Any litigation should expect aggressive §102/§103/§101 challenges.
- **This patent's claims center on a gesture causing a context-aware app launch.** Products that merely *transfer files* (most "share" features) do not infringe Claim 1 unless they also auto-launch the corresponding app and resume the matching activity. The presence or absence of the "[1d] execute an application based on the information" and "[1e] perform a function associated with the activity" steps is the swing factor in most rows of the ranking table.